

# Dynamics of continuous systems – Torsional vibrations of beams

## 1 Torsional vibrations of beams: assumptions

We will treat the problem of axial beam vibrations under the following assumptions:

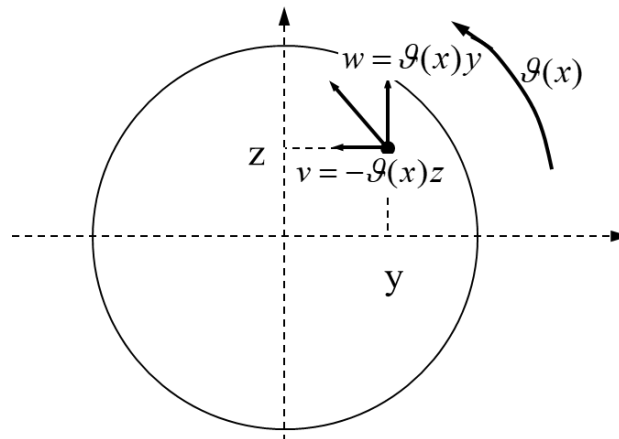
- 1) Small displacements.
- 2) Linear elastic constitutive law.
- 3) constant section and homogeneous material.
- 4) Damping effects can be neglected.
- 5) No forces are applied on the beam, except at the boundaries.
- 6) The beam section possesses a polar symmetry (i.e. any line passing through the centre of the section is a symmetry axis for the section).

## 2 Torsional deformation of a beams with section having polar symmetry

We choose, as usual, a cartesian reference with axis  $x$  coinciding with the axis of the beam. Due to the polar symmetry of the section, the choice of the  $y$  and  $z$  axes of the reference is arbitrary and has no effect on the theory presented below.

In case of polar symmetry of the beam's section, torsional deformation consists of a rigid rotation of each section around the  $x$  axis, described by the single scalar function  $\vartheta(x)$  defining the rotation of a generic section located at abscissa  $x$ .

Figure 2.1 shows the displacements of a generic point  $P$  located at coordinates  $x, y, z$  in the undeformed volume of the beam.



**Figure 2.1** Scalar components of displacements

The scalar components of the displacement vector are:

$$\begin{aligned}
 u &= 0 \\
 v &= -z\vartheta(x) \\
 w &= y\vartheta(x)
 \end{aligned}
 \tag{2.1}$$

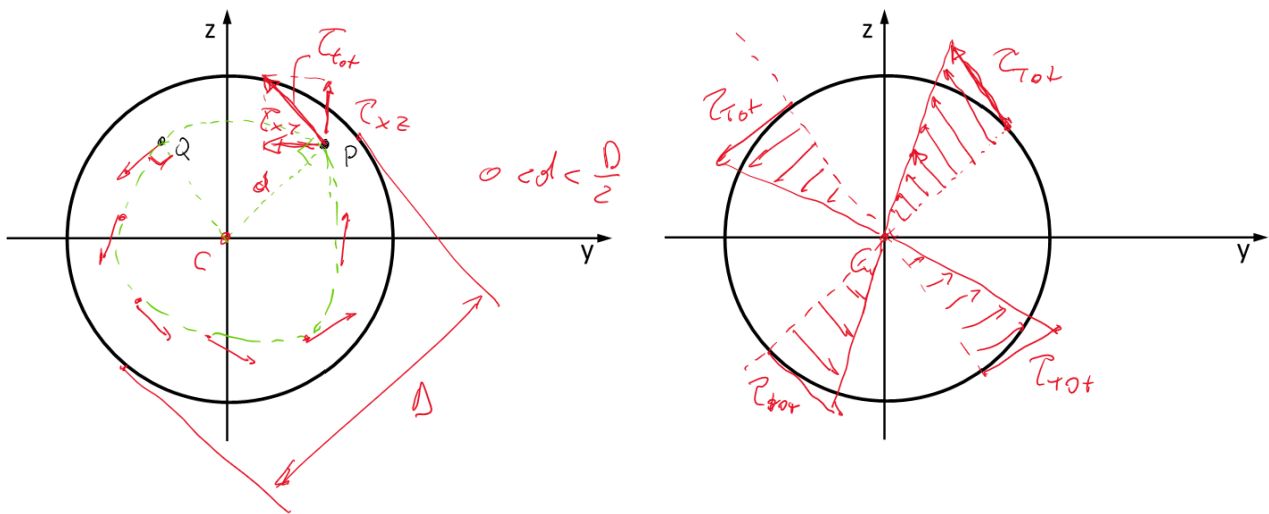
From the above expressions, the 6 independent components of strain can be obtained as follows:

$$\begin{aligned}\varepsilon_x &= \frac{\partial u}{\partial x} = 0 \quad ; \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = -z \frac{\partial g}{\partial x} \\ \varepsilon_y &= \frac{\partial v}{\partial y} = 0 \quad ; \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = 0 \\ \varepsilon_z &= \frac{\partial w}{\partial z} = 0 \quad ; \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z} = y \frac{\partial g}{\partial x}\end{aligned}\tag{2.2}$$

and then, from the linear elastic constitutive law, the normal stress  $\sigma_x$  is obtained:

$$\begin{aligned}\sigma_x &= \sigma_y = \sigma_z = 0 \\ \tau_{xy} &= -Gz \frac{\partial \vartheta}{\partial x} \\ \tau_{xz} &= Gy \frac{\partial \vartheta}{\partial x} \\ \tau_{yz} &= 0\end{aligned}\tag{2.3}$$

Eq. (2.3) shows that the torsional deformation of the beam leads to a distribution of tangential stresses on the section (normal  $x$ ) in both  $y$  and  $z$  directions. The vector sum of the two scalar components of shear stress in the section points in circumferential direction and the magnitude of the total shear stress at a given point in the volume of the beam is proportional to the distance of the point from the centre of the section, as shown in Figure 2.2.



**Figure 2.2: direction and magnitude of the total shear stresses produced by a torsional deformation applied to the beam with circular section**

The total axial force acting on a section of the beam is zero because due to the zero value of the normal stress  $\sigma_x$  at all points in the section, see Eq. (2.3). For the same reason, both components of bending moment are zero, hence it is confirmed the de-coupling of torsion and bending and stated when dealing with bending deformation of the beam. Furthermore, the shear forces along the y and z directions are both zero because the distribution of shear stresses along y and along defined by Eq. (2.3) are both anti-symmetric, whereas the geometry of the section is symmetric with respect to both axes.

Therefore, the only non-zero component of internal force is a moment  $M_x$  around axis x, which is called the *torsional moment* and often denoted as  $M_t$ . This is given by:

$$M_t = \int_A (-\tau_{xy}z + \tau_{xz}y) dA = G \int_A (z^2 + y^2) dA \frac{\partial \theta}{\partial x} = G I_p \frac{\partial \theta}{\partial x} \quad (2.4)$$

We see from Eq. (2.4) that the torsional moment is proportional to the gradient of the rotation the section  $\frac{\partial \mathcal{G}}{\partial x}$  through a constant which depends on material properties (through the shear modulus of elasticity  $G$ ) and on the shape and size of the section through parameter  $I_p$  called the *polar area moment of inertia of the section*. This term is defined as:

$$I_p = \int_A (y^2 + z^2) dA \quad (2.5)$$

and its value in SI units is expressed in  $m^4$ , like for the  $I_y$  area moment of inertia defined for bending. For a solid circular section (i.e. a section without a central hole), the polar area moment of inertia is given by:

$$I_p = \frac{\pi D^4}{32} \quad (2.6)$$

whilst for a hollow section with  $D_e$  and  $D_i$  the external and internal diameters the expression of  $I_p$  is:

$$I_p = \frac{\pi (D_e^4 - D_i^4)}{32} \quad (2.6b)$$

The case of a beam having a different shape of the section is not treated here for the sake of brevity, but we mention that in this case the relationship between the torsional moment  $M_t$  and the gradient of rotation is expressed by:

$$M_t = G I_T \frac{\partial \mathcal{G}}{\partial x} \quad (2.7)$$

where  $I_T$  is a different parameter depending on the size and shape of the section, which can be obtained from different formulae that can be found in handbooks, depending on the specific shape of the section

### 3 Equation of motion of torsional vibrations for a beam with section having polar symmetry

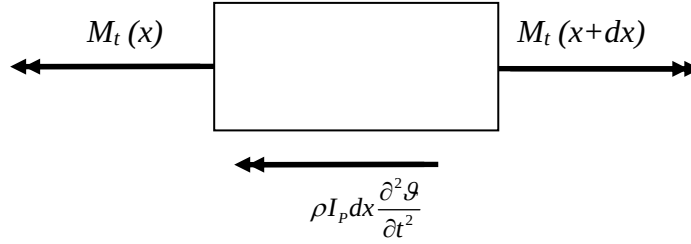
To represent the torsional movement of the beam, we consider function  $\mathcal{G}(x, t)$  representing the rotation around the  $x$  axis of the beam section as function of the space coordinate  $x$  and of time  $t$ . Then, we consider an infinitesimal piece of beam and represent the moments acting on it (no resulting linear force is generated in this case). These are:

- the torsional moment  $M_t(x)$  acting on the left face of the element;
- the torsional moment  $M_t(x+dx)$  acting on the right face of the element;
- the moment of the inertia forces generated on the considered piece of the beam, which is given by the following expression:

$$\rho I_p dx \frac{\partial^2 \mathcal{G}}{\partial t^2}$$

with  $\rho$  the material density and  $I_p$  the polar moment of inertia of the section.

The three moments listed above are represented in Figure 3.1 using the “double-arrow vector” representation, meaning that the moment acts along the rotational direction defined by the right hand when the thumb is pointing the direction of the double arrow.



**Figure 3.1:** Equilibrium of a beam element with infinitesimal dimension undergoing torsional vibration.

The application of D'Alembert's principle brings:

$$-M_t(x) + M_t(x+dx) = \rho I_p dx \frac{\partial^2 \vartheta}{\partial t^2} \quad (3.1)$$

With trivial intermediate steps we get:

$$\frac{\partial M_t}{\partial x} = \rho I_p \frac{\partial^2 \vartheta}{\partial t^2} \quad (3.2)$$

recalling Eq. (2.7), for a generic shape of the beam's section, we get:

$$\frac{\partial M_t}{\partial x} = G J_T \frac{\partial^2 \vartheta}{\partial x^2} \quad (3.3)$$

and therefore eq. (3.2) becomes:

$$G J_T \frac{\partial^2 \vartheta}{\partial x^2} = \rho I_p \frac{\partial^2 \vartheta}{\partial t^2} \quad (3.4)$$

which represents the equation of motion for the torsional vibrations of the beam. This equation may be rewritten in the form:

$$\frac{\partial^2 \vartheta}{\partial t^2} = c^2 \frac{\partial^2 \vartheta}{\partial x^2} \quad (3.5)$$

with  $c$  the speed of propagation of a wave of torsional deformation in the beam, which is given by:

$$c = \sqrt{\frac{G J_T}{\rho I_p}} \quad (3.6)$$

In case the section of the beam has polar symmetry, which is the focus of this note,  $J_T = I_p$  and the speed of propagation becomes only dependent on material properties  $G$  and  $\rho$ .

Propagative and stationary solutions for equation (3.5) are the same as those derived for the axial vibration of a beam. Moreover, the prescription of boundary conditions to the stationary solution of eq. (3.5) and the derivation of the system's natural frequencies and modes of vibrations are also carried out in the same way as for the cases previously examined in detail, to which the reader is referred.